

Splunk Detection Lab

Detecting and Analyzing Alerts

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Date	September 3, 2026
Lab Environment	Ryzen 7 Pro 5850U, 16GB RAM, 512GB NVMe SSD
Hypervisor	Oracle VirtualBox
VM OS	Ubuntu Server 22.04 (Splunk Enterprise Head)
VM OS	Windows 10 (Monitored Endpoint)
Report Version	v1.0

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1. Executive Summary

The key objective of this project is to mimic SOC monitoring using Splunk Security Information and Event Management (SIEM) Tool. The Splunk Enterprise was downloaded on Ubuntu server virtual machine (Splunk Indexer and Search Head). The Splunk Universal Forwarder was installed on Windows 10 VM to forward Windows event logs to Indexer/Search Head. At endpoint, inputs.conf file was created to enable telemetry of Windows Security, Application, and System logs. All these events were indexed to a custom index '**windows**' in Splunk. In Splunk, **Splunk Add-on for Microsoft Windows** was installed for better parsing of event logs. In Splunk, queries were tested to detect, summarize, filter, and tabularize Windows events like failed login attempts (**4625**), new account created (**4720**), and process created (**4688**). By utilizing these queries, alerts were created for password brute-force attack, new-user creation, and suspicious process creation. A customized SOC monitoring dashboard was created for real-time detection of alerts. This lab project gave hands-on experience of deployment and configuration of SIEM and forwarder, querying specific events, and transforming useful fields into statistics/visualizations that can assist a SOC analyst to detect and respond to alerts.

Key Achievements

- Learned deployment and configuration of Splunk Indexer/Search Head and Universal Forwarder.
- Learned how the "Splunk Add-on for Microsoft Windows" improves event log parsing and field extraction.
- Learned how to query specific events, extract statistics, and tabularize fields.
- Created customized alerts for brute-force, new-user creation, and suspicious PowerShell command execution.
- Learned how to create dashboards for visual representation of alerts and metrics.

2. Lab Architecture & Environment

2.1 Hardware

Component	Specification
Laptop Model	Lenovo ThinkPad T14
Processor	AMD Ryzen 7 Pro 5850U (8 Cores / 16 Threads)
RAM	16GB
Storage	512GB NVMe SSD
OS	Windows 11

2.2 Software Stack

Tool	Purpose
VirtualBox	Hypervisor for VM hosting
Windows Guest (Win 10)	Monitored Endpoint
Ubuntu Server (Guest)	To Install Splunk Indexer and Search Head
Splunk Enterprise	Indexer and Search Head
Splunk Universal Forwarder	Events Logs Forwarder
Splunk Add-on for Microsoft Windows	Add-on for better Logs Parsing

2.3 Splunk Installation

To use Splunk Enterprise for free (**for 60 days only**), created account on splunk.com, and waited for verification of email that took almost 8 hours (can take up to 24 hours).

After email verification, logged on splunk.com and download Splunk Enterprise trial version for Linux Debian based distributions (.deb) on Ubuntu Server (opened in Windows 10 VM using ssh) using wget command.

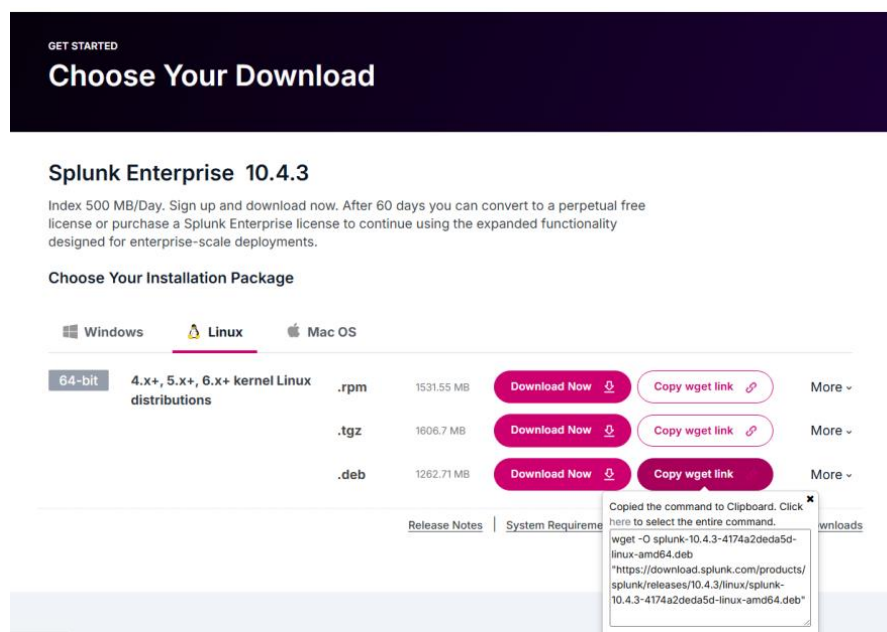


Figure 1: Splunk Enterprise Download

Installed the downloaded .deb package,

```
sudo dpkg -i splunk-*.deb
```

And created an administrative user (user name: splunk) during installation and also set the password.

After installation, started Splunk

```
sudo /opt/splunk/bin/splunk start
```

This gave error that Splunk can't be started as root. So changed the ownership of Splunk to splunk user,

```
sudo chown -R splunk:splunk /opt/splunk
```

Then started Splunk

```
sudo -u splunk /opt/splunk/bin/splunk start
```

For boot-start of Splunk,

```
sudo -u splunk /opt/splunk/bin/splunk enable boot-start -user splunk
```

Checked status of Splunk,

```
sudo /opt/splunk/bin/splunk status
```

Accessed Splunk-UI in Windows VM browser at <http://<Ubuntu Server IP>:8000>

First of all, navigated to settings (in the top bar) → searched for and clicked 'Forwarding and receiving' → Configure receiving → Clicked 'New Receiving Port' and set 9997 as receiving port. Data from forwarder will be received on this port.

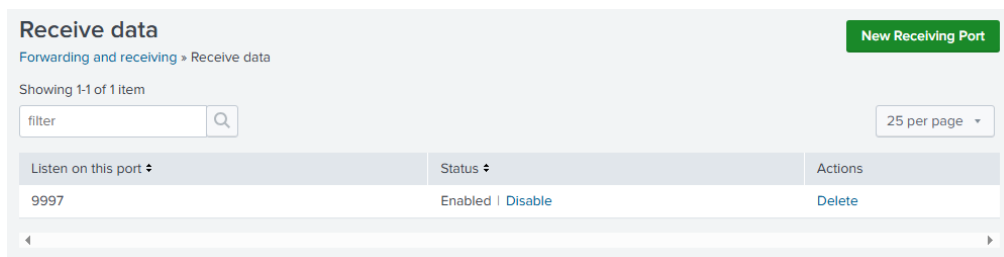


Figure 2: Set 9997 as Receiving Port

After that, again navigated to settings (in the top bar) → searched for and clicked 'Indexes' → New Index → created a new index 'windows' with all default settings. All logs from Windows endpoint were indexed in this index.

2.4 Universal Forwarder Installation

Logged on splunk.com with same credentials and downloaded Splunk Universal Forwarder for 64-bits Windows operating system.

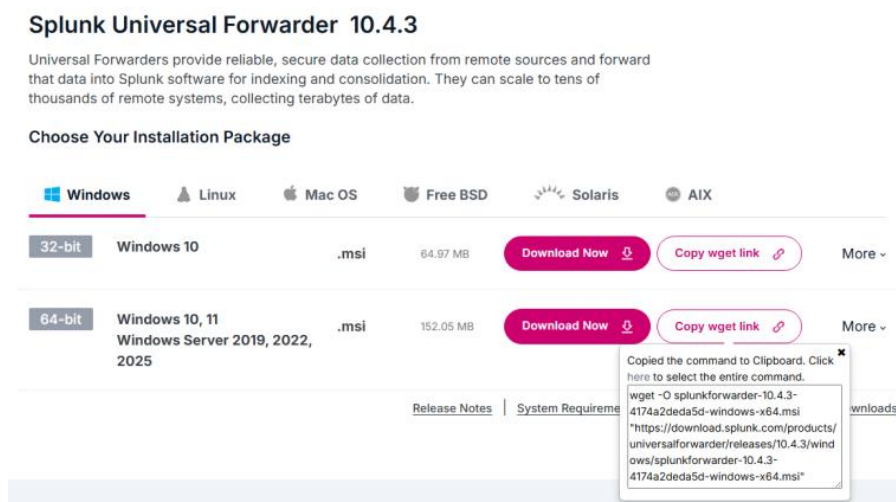


Figure 3: Universal Forwarder Download

During installation, provided Ubuntu server IP as Indexer IP address and set data receiving port as 9997.

Started universal forwarder from cmd,

```
net start SplunkForwarder
```

Status of forwarder was checked using following command in cmd,

```
sc query SplunkForwarder
```

2.5 Events Search and Creating Alerts

In Splunk-UI, navigated to Search & Reporting option, and searched for index = "windows", and this fetched the forwarded events from Windows 10 endpoint.

For failed login attempt, following query was used,

```
index="windows" EventCode = 4625
```

To detect repeated failed login attempts, 4625 event logs were bucketed by time,

```
index="windows" EventCode = 4625 | bucket _time span=5m | stats count by user, _time
```

And if the count of failed logins was greater than 3, password brute-force attack alert would be triggered. This maps to MITRE ATT&CK credential access technique T1110.001 (password guessing brute-force attack).

```
index="windows" EventCode = 4625 | bucket _time span=5m | stats count by user, _time | where count>3
```

Next, new user account creation was detected by event 4720, after creating a new user account.

```
index="windows" EventCode = 4720 | table _time, src_user, user
```

src_user is the user who created the new user account, and user is the newly created account. Creation of a new local account is mapped to MITRE ATT&CK technique T1136.001 for persistence.

After that, alert was created for detection of MITRE ATT&CK technique T1059.001, which is abuse of PowerShell command and scripting interpreter for execution.

```
index="windows" EventCode = 4688 Process_Command_Line = *Encoded* | table  
_time, parent_process_name, process_name, Process_Command_Line
```

The above alert was triggered by running following command in cmd,

```
powershell.exe -NoProfile -EncodedCommand  
VwByAGkAdABlAC0ASABvAHMAAdAAgACIAUwBwAGwAdQBwAGsAIAB0AG  
UAcwB0ACAAZQB2AGUAbgB0ACAANAA2ADgAOAAiAA==
```

For telemetry of process creation events and log command line process creation, mandatory settings are following,

In Group Policy → Windows Settings → Security Settings → Advanced Audit Policy Configuration → System Audit Policies - Local Group Policy → Detailed Tracking → Audit Process creation (Tick Success)

In Group Policy → Administrative Templates → System → Audit Process Creation → Enable “Include command line in process creation events”.

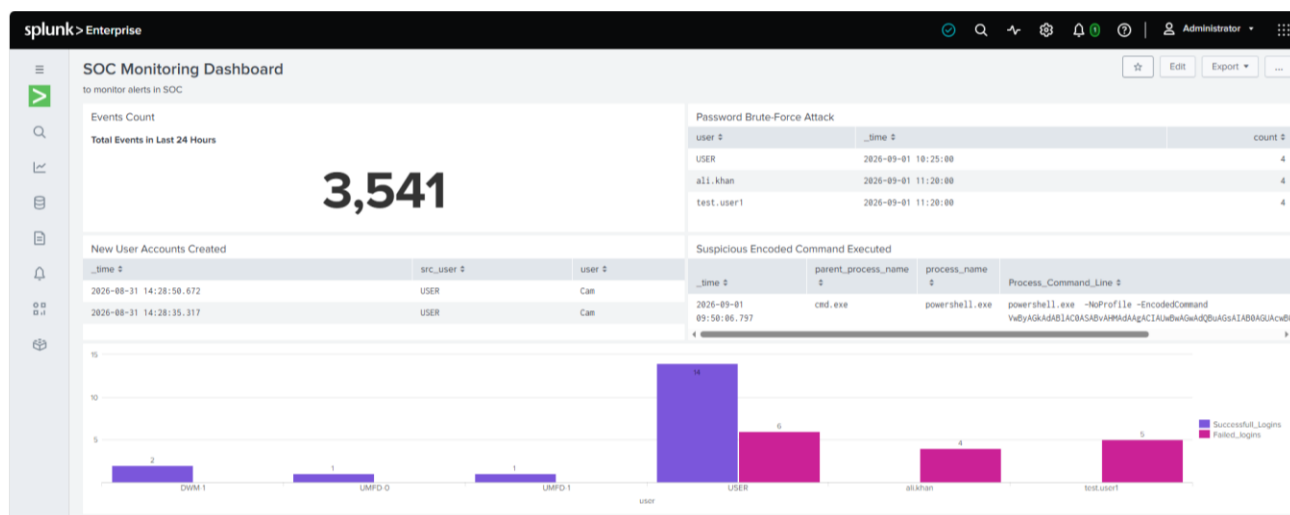
2.6 SOC Monitoring Dashboard

Again navigated to Search & Reporting → Dashboards (in the left menu) → Create a New Dashboard (Entered Title and Selected Classic Dashboard)

Added following 5 panels in dashboard,

- Single value panel to display “Total events in last 24 hours” in the ‘windows’ index.
- Statistics table to display brute-force attack alerts in last 24 hours, using brute-force attack detection query.
- Statistics table to display new user creation alerts in last 24 hours, using new-user creation detection query.
- Statistics table to display suspicious PowerShell command execution alerts in last 24 hours, using PowerShell command execution detection query.
- Column chart was used to display failed and successful logins by all users in the last 24 hours, using the following query,

```
index="windows" (EventCode = 4624 OR EventCode = 4625) | where Logon_Type  
IN (2,7, 10,11) | stats count(eval(EventCode = 4624)) as Successfull_Logins  
count(eval(EventCode = 4625)) as Failed_logins by user
```



3. Splunk vs. Elastic Stack

Elastic Stack	Splunk
Elastic Stack is a general purpose data collection, search, and analysis tool; it has a security module (Elastic Security) to monitor network devices and collect event logs from them.	Splunk is also a general purpose data collection, search, and analysis tool; it offers security monitoring, alert detection, and response through add-ons like Splunk Enterprise Security.
Components of Elastic Stack have to be installed separately, i.e., Elasticsearch (storage/indexing engine), Logstash (data processing/ingestion), and Kibana (dashboard).	Splunk Enterprise comes in a single package containing indexer, search head, and user web interface dashboard.
After installation of Elasticsearch and Kibana, Kibana has to be enrolled with Elasticsearch.	No need to enroll web interface.
Elastic Agent is installed on the endpoint using a PowerShell command to add it as an agent.	Splunk Universal Forwarder is installed on the endpoint to add it as an agent.
Agent policy is created with integrations for telemetry of event logs.	inputs.conf is used to configure event log telemetry.
Custom detection rules can be created using the GUI.	SPL (Search Processing Language) queries are used to create alerts.

4. Conclusion

Network visibility is a key component of cybersecurity, and Splunk is one of the widely used tools for this purpose. In this lab, Splunk was installed and configured on Ubuntu Server, and a Windows 10 endpoint was enrolled. Telemetry parameters were configured using the `inputs.conf` file at the monitored endpoint, with group policy modified to enable detailed process creation monitoring. Windows Application, Security, and System event logs were forwarded to Splunk, queried using SPL, and used to create custom alerts.

Alerts were created for detection of credential access with password guessing brute-force (T1110.001), persistence using creation of new local user account (T1136.001), and execution using PowerShell command and scripting interpreter abuse (T1059.001). A customized SOC dashboard was created for real-time monitoring of these alerts. Monitoring tool installation and configuration, event telemetry adjustment, impact of better log parsing, event searching, customized detection alert creation, and dashboard development were the key skills learned in this lab that are pivotal for a junior SOC analyst.